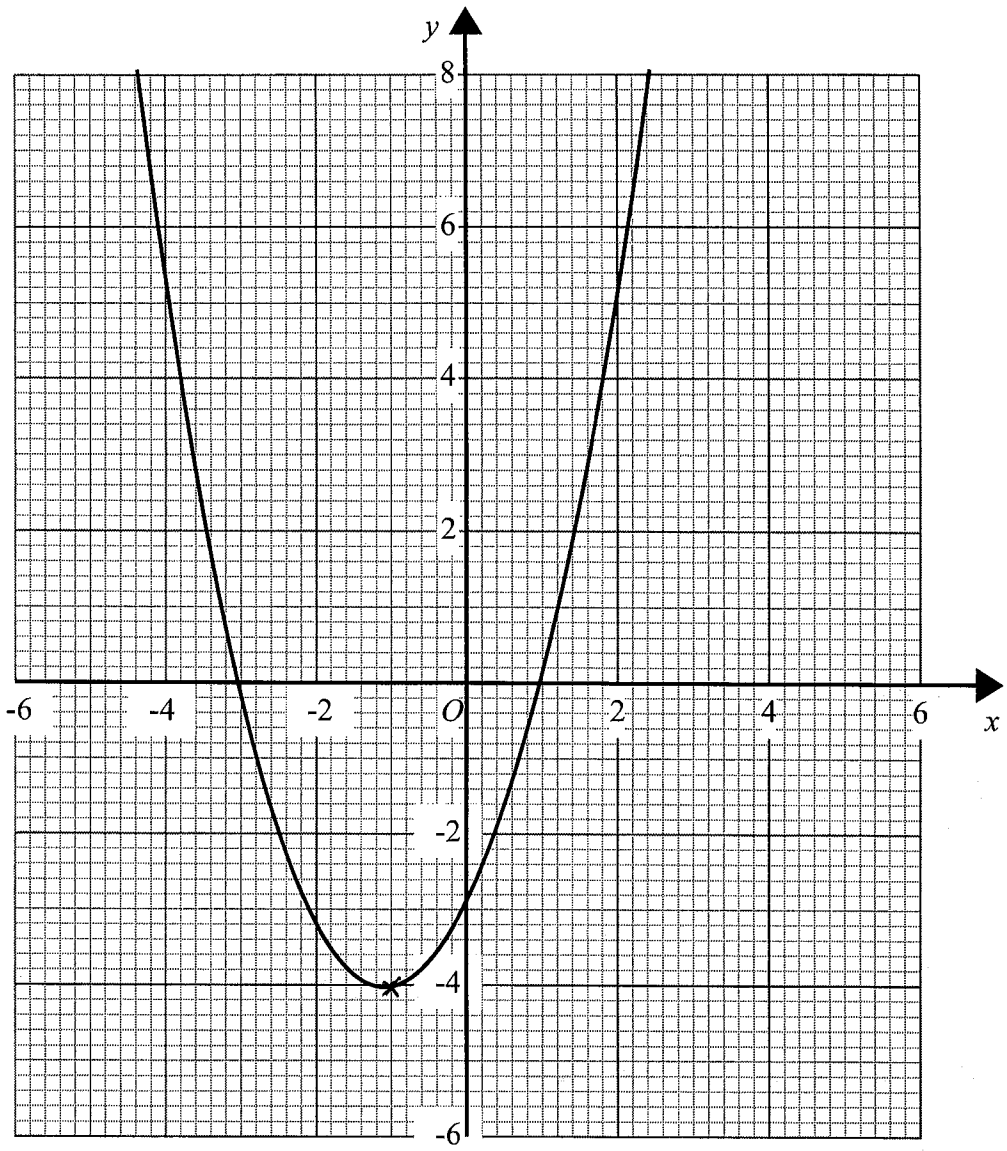
1 Here is the graph of $y = x^2 + 2x - 3$



(a) Write down the turning point of the graph $y = x^2 + 2x - 3$

(.....−1..... , ...−4.....)
(1)

(b) Use the graph to find the roots of the equation $x^2 + 2x - 3 = 0$

$x = 1$ and $x = -3$
(2)

(Total for Question 1 is 3 marks)

2 (a) Write 8.9×10^{-2} as an ordinary number.

0.089

(1)

(b) Work out the value of $(4.6 \times 10^5) \times (7.5 \times 10^3)$
Give your answer in standard form.

3450000000

3.45×10^9

(2)

(Total for Question 2 is 3 marks)

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3 In a sale, the normal price of a TV is reduced by 15%.
The sale price of the TV is £595

Work out the normal price of the TV.

$x \times 0.85 = 595$

$x = \frac{595}{0.85}$

$= 700$

£ 700

(Total for Question 3 is 2 marks)

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4 (a) Expand and simplify $(5p - 4)(p - 2)$

$5p^2 - 10p - 4p + 8$

$5p^2 - 14p + 8$

(2)

(b) Factorise $a^2 + 6a - 55$

$(a + 11)(a - 5)$

(2)

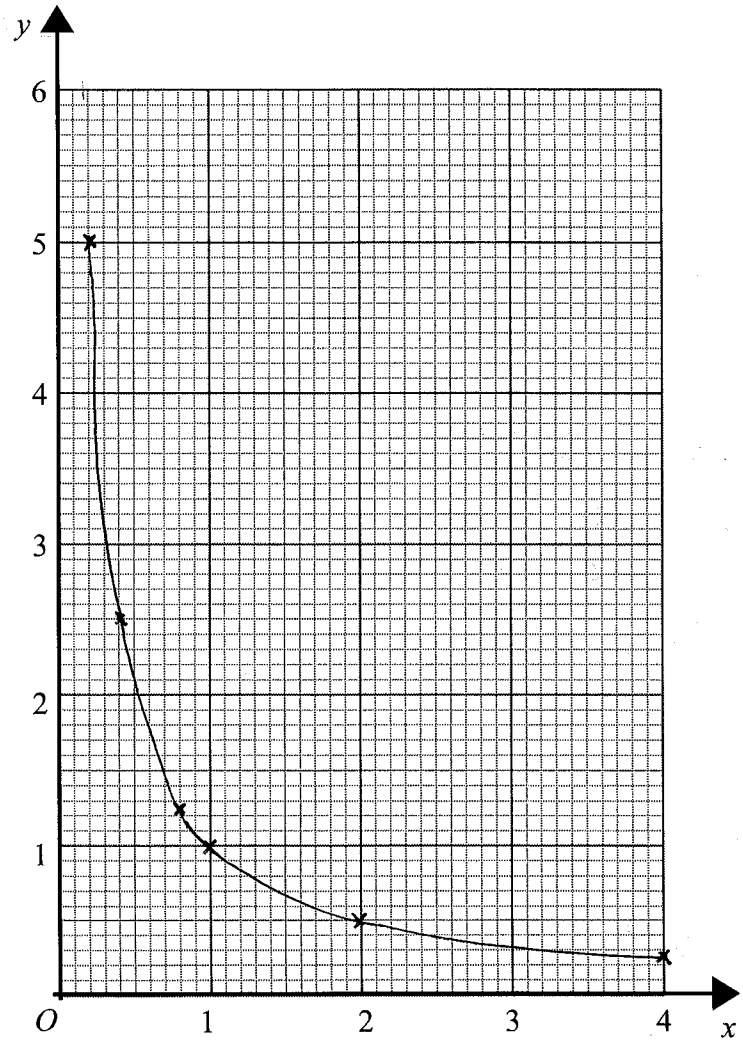
(Total for Question 4 is 4 marks)

5 (a) Complete the table of values for $y=\frac{1}{x}$

x	0.2	0.4	0.8	1	2	4
y	5	2.5	1.25	1	0.5	0.25

(2)

(b) On the grid, draw the graph of $y=\frac{1}{x}$

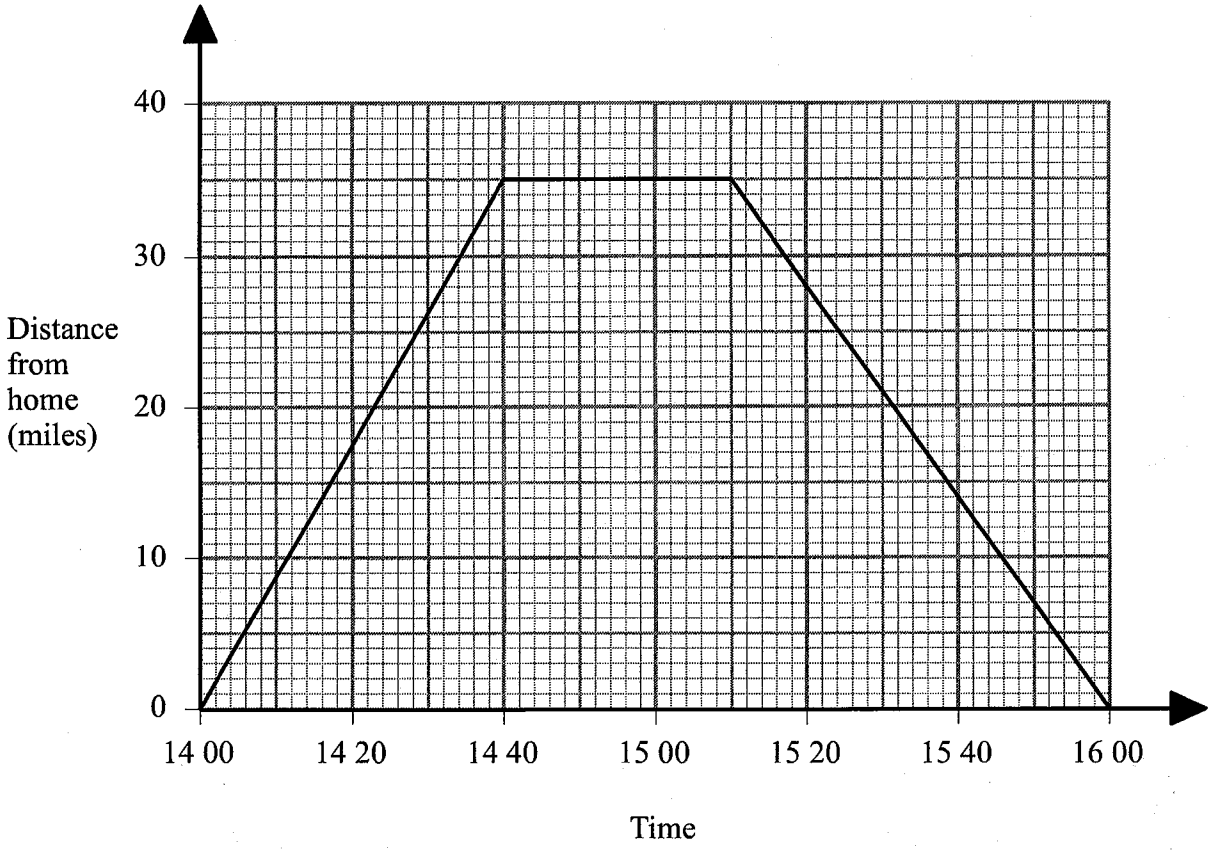


(2)

(Total for Question 5 is 4 marks)

6 Emily drove to the beach. She stayed at the beach and then she drove back home.

Here is Emily's travel graph.



(a) For how many minutes did Emily stay at the beach?

1440 to 1510

30 minutes
(1)

(b) What was Emily's average speed on her **journey to the beach**?

35 miles in 40 mins

$$\text{speed} = \frac{\text{distance}}{\text{time}} = \frac{35}{2/3}$$

52.5 miles/hour
(2)

(Total for Question 6 is 3 marks)

7 Solve $b^2 - 8b + 12 = 0$

$(b-2)(b-6) = 0$

$b = 2 \quad b = 6$

$b = 2$ or $b = 6$

(Total for Question 7 is 3 marks)

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8 Liquid **A** has a density of 1.1 g/ml

150 ml of Liquid **A** is mixed with some of Liquid **B** to make Liquid **C**.

Liquid **C** has a mass of 378 g and a density of 1.05 g/ml

Find the density of Liquid **B**.

Give your answer correct to 3 significant figures.

$\text{density} = \frac{\text{mass}}{\text{volume}}$

A: $D = 1.1$, $V = 150$

$\text{Mass} = d \times V$
$= 1.1 \times 150$
$= 165\text{g}$

C: $m = 378$, $d = 1.05$

$\text{volume} = \frac{m}{d}$
$= \frac{378}{1.05}$
$= 360\text{ ml}$

B: $\text{mass} = 378 - 165$
$= 213\text{g}$

$\text{volume} = 360 - 150$
$= 210\text{ ml}$

$\text{density} = \frac{213}{210} = 1.01$

1.01 g/ml

(Total for Question 8 is 3 marks)